

SEAMLESS DEPARTMENT EVENT BOOKING APP

School of Computing

**Bachelor of Technology
in
Computer Science & Engineering
By**

K.Bunny Preetham chary (VTU26337)

*10211CS224
Full Stack Application Development
Slot : E2*



**DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
SCHOOL OF COMPUTING**

**VEL TECH RANGARAJAN Dr.SAGUNTHALA R&D
INSTITUTE OF SCIENCE AND TECHNOLOGY
(Deemed to be University Estd u/s 3 of UGC Act, 1956)
CHENNAI 600 062, TAMILNADU, INDIA**

May, 2026

BONAFIDE CERTIFICATE

It is certified that the work contained in the Full Stack Application Development report titled "SEAMLESS DEPARTMENT EVENT BOOKING APP " by K.Bunny Preetham chary (VTU26337) has been carried out in Winter semester of Academic year 2025-2026(WS2526).

Signature of Faculty

Dr. S.Hemamalini

Assistant Professor(Senior Grade)

Computer Science & Engineering

School of Computing

Vel Tech Rangarajan Dr.Sagunthala R&D

Institute of Science and Technology

May, 2026

Signature of Course Co-ordinator Sign

Dr.M.Sumitha

Assistant Professor (Senior Grade)

Computer Science & Engineering

School of Computing

Vel Tech Rangarajan Dr.Sagunthala R&D

Institute of Science and Technology

May, 2026

DECLARATION

I declare that this written submission represents my ideas in my own words and where others' ideas or words have been included, we have adequately cited and referenced the original sources. I also declare that i have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in our submission. I understand that any violation of the above will be cause for disciplinary action by the Institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

K.Bunny preetham chary

Date:06/05/2026

ABSTRACT

The Seamless Department Event Booking App is a web-based system designed to simplify and automate the process of booking departmental events, venues, and resources. In many institutions, event booking is handled manually, which often leads to scheduling conflicts, double bookings, and inefficient communication. This application addresses these challenges by providing a centralized and user-friendly platform.

The system allows students, faculty, and administrators to view available venues, check real-time schedules, and book events and efficiently. It uses a dynamic booking mechanism to ensure that no two users can reserve the same resource at the same time, thereby preventing conflicts and improving reliability..

Keywords: React JS, Event Booking, Ticket Reservation, Web Application, Department Events, Real-Time Booking, Input Validation, User Interface, Event Management, Online Ticketing.

LIST OF FIGURES

1.1	Framework	4
3.1	Execution Procedure	9
5.1	User Login Page for Event Booking System	17
5.2	Display of Upcoming Department Events	17
5.3	Seat Selection Interface for Technical Fest 2026	18
5.4	Customer Details and Payment Summary Page	18
5.5	Ticket Booking Confirmation Screen	19
5.6	Final Ticket Details with Seat and Payment	19

LIST OF TABLES

5.1	Example Comparison	17
8.1	Mapping of Project to Complex Engineering Problem (CEP)	26
8.2	Mapping of Project to Complex Engineering Activities (CEA)	27
8.3	SDG Mapping for the Project	28

LIST OF ACRONYMS AND ABBREVIATIONS

API	Application Programming Interface
CI	Continuous Integration
CSS	Cascading Style Sheets
HTML	HyperText Transfer Markup Language
IDE	Integrated Development Environment
JS	JavaScript
JSX	JavaScript XML
NPM	Node Package Manager
ORM	Object Relational Mapping
QR	Quick Response
UI	User Interface
URL	Uniform Resource Locator

TABLE OF CONTENTS

	Page.No
ABSTRACT	iii
LIST OF FIGURES	v
LIST OF TABLES	vi
LIST OF ACRONYMS AND ABBREVIATIONS	vii
1 INTRODUCTION	1
1.1 Introduction	1
1.2 Aim of the Project	2
1.3 Scope of the Project	2
1.4 Framework	3
2 SURVEY OF SIMILAR APPLICATION IN MARKET	5
3 FRAMEWORK DESCRIPTION	7
3.1 Frontend Implementation	7

3.2	Backend Implementation	8
3.3	Database Implementation	10
4	METHODOLOGIES	12
4.1	Project Architecture	12
4.2	DataFlow Diagram	13
4.3	Module 1:Event Information Display Module	13
4.4	Module 2:Ticket Booking Management Module	13
4.5	Module 3:Validation and Confirmation Module	14
5	RESULTS AND DISCUSSIONS	16
6	CONCLUSION AND FUTURE ENHANCEMENTS	20
6.1	Conclusion	20
6.2	Future Enhancements	21
	References	21
7	SOURCE CODE	23
7.1	Source Code	23
CHAPTER 8	ADDRESSING COMPLEX ENGINEERING PROBLEM AND SDG	26
8.1	Mapping of Project to Complex Engineering Problem (CEP)	26

8.2	Mapping of the Project to Complex Engineering Activities(CEA)	27
8.3	SDG Mapping	28

References	28
-------------------	-----------

Chapter 1

INTRODUCTION

1.1 Introduction

The Seamless Department Event Booking App is a web-based application developed to streamline and automate the process of booking departmental events, venues, and related resources. Traditional manual booking systems often result in scheduling conflicts, inefficient communication, and lack of transparency. This system addresses these issues by providing a centralized platform that enables users to view availability, schedule events, and manage bookings in real time.

The application supports multiple users, including students, faculty, and administrators, allowing them to access updated schedules and make reservations accurately. It incorporates a dynamic allocation mechanism to prevent double bookings and ensure efficient utilization of resources. Additional features such as booking confirmation, event management, and a simulated payment module.

1.2 Aim of the Project

The main aim of this project is to develop a responsive and efficient web application that enables users to book tickets for departmental events online in a simple, secure, and organized manner. The system aims to automate the ticket booking process, reduce manual effort, prevent overbooking, and improve overall event management within educational institutions.

1.3 Scope of the Project

The scope of this project includes designing and developing a ticket booking platform specifically for internal departmental events. The application allows users to:

1. View event details such as date, time, venue, and ticket price.
2. Book tickets online by entering required details
3. Validate user inputs for accurate booking
4. Prevent booking beyond available ticket count
5. Display booking confirmation and total amount
6. Update ticket availability dynamically after each booking

Future enhancements may include database integration, payment gateway support, email notifications, and admin event management features.

1.4 Framework

The project is developed using the React JS framework, a widely used JavaScript library for building dynamic and interactive user interfaces. React allows the development of reusable components and efficient state management through hooks such as ‘useState‘, which helps in handling ticket availability and booking data dynamically. The application also utilizes JSX for writing component-based user interface code in a structured and readable format, CSS3 for designing a responsive and visually appealing layout, and Vite as the development and build tool to ensure faster project setup and improved performance during development. This combination of modern frontend technologies provides scalability, maintainability, high performance, and an enhanced user experience for the application.

1.4 Framework

The project “Instant Booking Platform for Department Events” is developed using the following technologies and tools:

Technology / Tool	Logo	Purpose	Role in Project
React JS		JavaScript library for building user interfaces	Used to build reusable UI components and manage application state.
JavaScript (ES6)		Programming language of the web	Used for implementing logic, state management and functionality.
HTML5		Markup language for structuring web pages	Used for structure and content organization.
CSS3		Style sheet language	Used for designing and styling the user interface.
Vite		Modern build tool	Used for fast development server and build process.
VS Code		Code editor	Used for writing and developing the code.

Figure 1.1: Framework

Chapter 2

SURVEY OF SIMILAR APPLICATION IN MARKET

Several ticket booking applications are currently available in the market for booking events, movies, concerts, seminars, and live shows. Popular platforms such as BookMyShow, Eventbrite, and Ticketmaster provide online ticket booking, event listing, payment processing, and booking confirmation features. BookMyShow is widely used in India for booking movie tickets, concerts, plays, sports, and other live events. Eventbrite allows organizers to create events, sell tickets online, and manage event registrations. Ticketmaster mainly focuses on concerts, sports, theatre, and large entertainment events.

These applications are highly useful for large-scale public events, but they may be too complex for small internal department events in colleges. They often include advanced features such as online payments, event promotion, ticket resale, organizer dashboards, and mar-

keting tools. In comparison, the proposed Instant Booking Platform for Department Events is simpler and specially designed for academic department events. It allows students and faculty to view event details, enter booking information, validate inputs, prevent overbooking, update ticket availability, and display booking confirmation instantly.

This project focuses on providing a lightweight, user-friendly, and easy-to-use booking system for seminars, technical fests, workshops, and department-level programs. Unlike commercial platforms, this application is developed mainly for educational purposes and internal use, making it suitable for small institutions and academic project demonstrations.

Chapter 3

FRAMEWORK DESCRIPTION

3.1 Frontend Implementation

3.1.1 Environment Setup

The frontend of the project is developed using **React JS** with Vite as the build tool. The development environment includes Node.js, npm, Visual Studio Code, and a web browser. **React JS** is used to create interactive user interfaces, while Vite provides fast project creation and quick execution. **CSS3** is used for designing the layout and improving the visual appearance of the application.

3.1.2 Structure of the Project

The project follows a simple component-based structure. The main application file is **App.jsx**, which manages the ticket booking logic and state. Separate components such as **EventDetails.jsx** and **BookingForm.jsx** are used to display event information and handle user booking input. The styling of the application is maintained in **App.css**.

This structure improves code readability, reusability, and maintenance.

3.1.3 Execution Procedure

To execute the project, the user must open the project folder in Visual Studio Code and install the required dependencies using **npm** install. After installation, the command **npm run dev** is used to start the development server. The application runs in the browser using the local URL provided by Vite, usually **http://localhost:5173/**. Users can then view event details, fill the booking form, and book tickets online.

3.2 Backend Implementation

3.2.1 Environment Setup

In this project, a separate backend server is not implemented because the application is designed as a frontend-based React JS project. The ticket booking logic, validation, ticket count update, and booking confirmation are handled directly within the React application using JavaScript and React state management.

3.2.2 Structure of the Project

The backend-like logic is managed inside the frontend components. The **useState** hook is used to store available tickets, form input values, error messages, and booking summary details. The booking function checks whether the entered details are valid and ensures that the num-

ber of tickets booked does not exceed the available ticket count.

3.2.3 Execution Procedure

Since there is no separate backend, no backend server execution is required. The entire application runs through the React development server. When the user submits the booking form, the logic is executed immediately in the browser, and the updated ticket availability and booking confirmation are displayed on the same page.

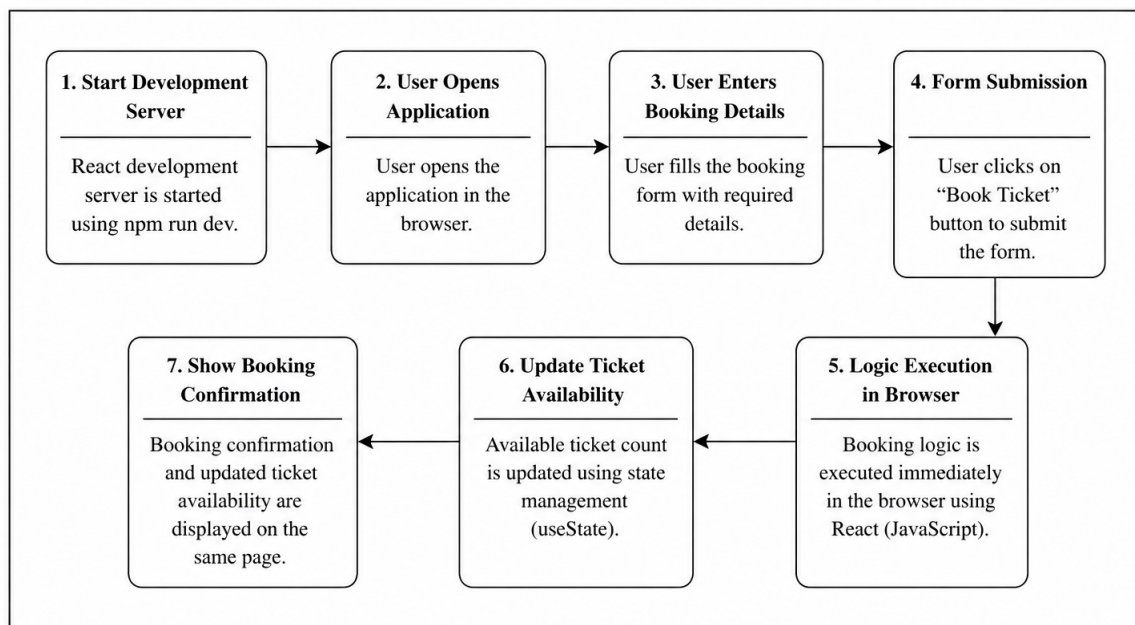


Figure 3.1: Execution Procedure

Figure 3.1: Execution Procedure

3.3 Database Implementation

3.3.1 Database Configuration

A physical database is not used in this project. The application stores data temporarily using React state. Ticket availability, user input, and booking summary are maintained only during the current browser session. This approach is suitable for a basic academic project and demonstration purpose.

3.1.2 Schema Structure

Since no database is connected, there is no permanent schema structure. However, the application uses a logical data structure to manage event and booking details. The event data includes event name, department name, date, time, venue, ticket price, and available tickets. The booking data includes user name, email ID, department, number of tickets, and total amount.

3.1.3 Tables created

No database tables are created in the current version of the project. For future enhancement, tables such as **Event Table**, **User Table** and **Booking Table** can be created to store event details, user information, and ticket booking records permanently. This would allow the system to maintain booking history and support multiple events.

Chapter 4

METHODOLOGIES

4.1 Project Architecture

The Seamless Department Event Booking App follows a three-tier architecture consisting of the presentation layer, application layer, and data layer. The presentation layer provides the user interface through which users can view events, check availability, and make bookings. The application layer handles the core logic of the system, including processing requests, managing bookings, and preventing scheduling conflicts. The data layer is responsible for storing and managing information such as user details, event schedules, and booking records.

The system operates by sending user requests from the frontend to the backend, where availability is verified and the booking is processed. Once confirmed, the details are stored in the database, and a confirmation is returned to the user.

4.2 DataFlow Diagram

The data flow of the system begins when the user accesses the event booking page and views the event details. The user then enters booking information including name, email, department, and number of tickets. This data is passed to the validation module, where input fields are checked for correctness. If validation is successful, the booking module processes the request, updates ticket availability, and generates the booking summary. Finally, the confirmation module displays the booking details and total amount to the user.

4.3 Module 1:Event Information Display Module

The Event Details Module is responsible for displaying complete information about the department event. It shows the event name, department name, date, time, venue, ticket price, and currently available tickets. This module helps users understand the event details before making a booking decision.

4.4 Module 2:Ticket Booking Management Module

The Ticket Booking Module allows users to enter booking details such as name, email ID, department, and required number of tickets.

It handles form submission and sends the entered data to the validation and booking logic for further processing.

4.5 Module 3: Validation and Confirmation Module

This module validates all user inputs to ensure mandatory fields are filled, email format is correct, and ticket quantity is positive and within available limits. Once validation succeeds, the system updates the ticket count and displays the booking confirmation summary including user name, event name, tickets booked, and total amount.

Advantages of this Project

a) User-Friendly Interface:

The application provides a simple, clean, and interactive user interface that enables students and faculty members to book event tickets easily without technical difficulty. Its intuitive design improves accessibility and enhances the overall user experience.

b) Real-Time Ticket Availability:

The system updates the available ticket count instantly after each successful booking. This real-time update mechanism helps prevent overbooking and ensures accurate ticket management.

c) Efficient Event Management:

The platform automates the ticket booking process, reducing manual work for event organizers and improving the efficiency of event registration and participant management.

Disadvantages

The current version of the application does not include backend or database integration; therefore, booking data is not stored permanently and will be lost upon page refresh or browser closure. Additionally, the system currently supports only single-event booking and does not provide advanced features such as payment gateway integration, user authentication, or multi-event management.

Chapter 5

RESULTS AND DISCUSSIONS

The Seamless Department Event Booking App was successfully developed and tested to evaluate its performance and usability in managing departmental events. The system effectively allows users to view available venues, check real-time schedules, and book events without conflicts. The dynamic booking mechanism ensures that duplicate bookings are avoided, which improves reliability and accuracy.

During testing, the application demonstrated smooth interaction between the frontend, backend, and database. Users were able to complete bookings, and the confirmation process, including simulated payment and email notifications, worked as expected. The system also showed good responsiveness and handled multiple user requests efficiently.

Table 5.1: Example Comparison

Applications	Advantages
Manual Booking System	Time-consuming and prone to errors
Offline Registration	Difficult to track real-time availability
Google Forms Registration	No instant ticket count update
Instant Booking Platform for Department Events	Real-time booking, validation, and instant confirmation

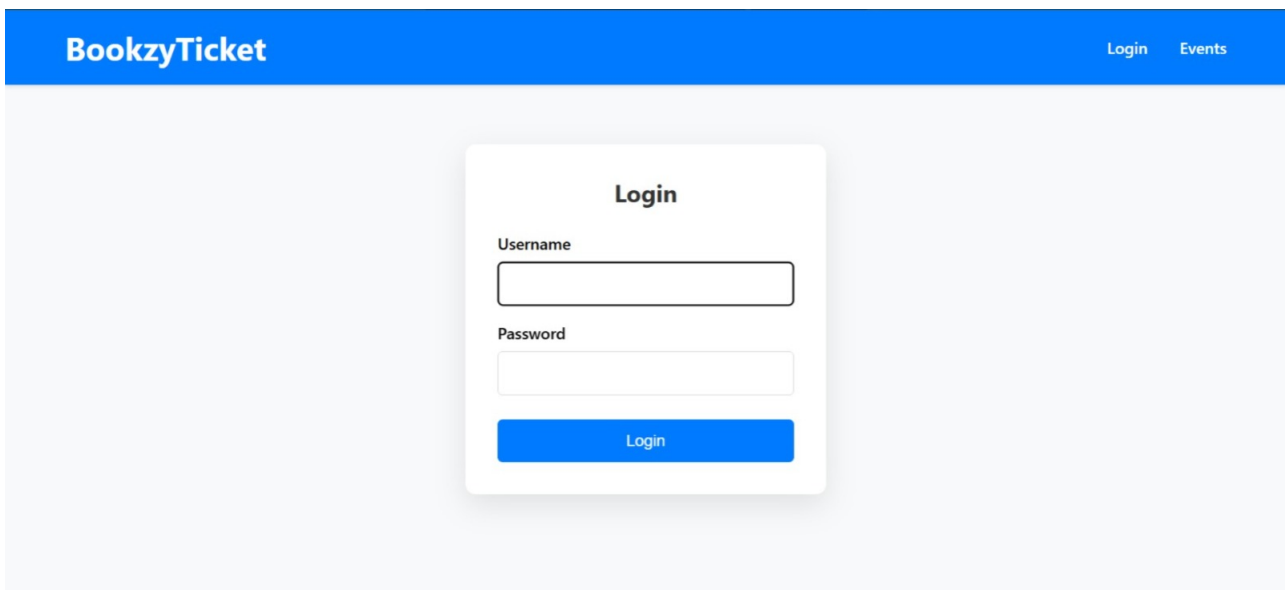
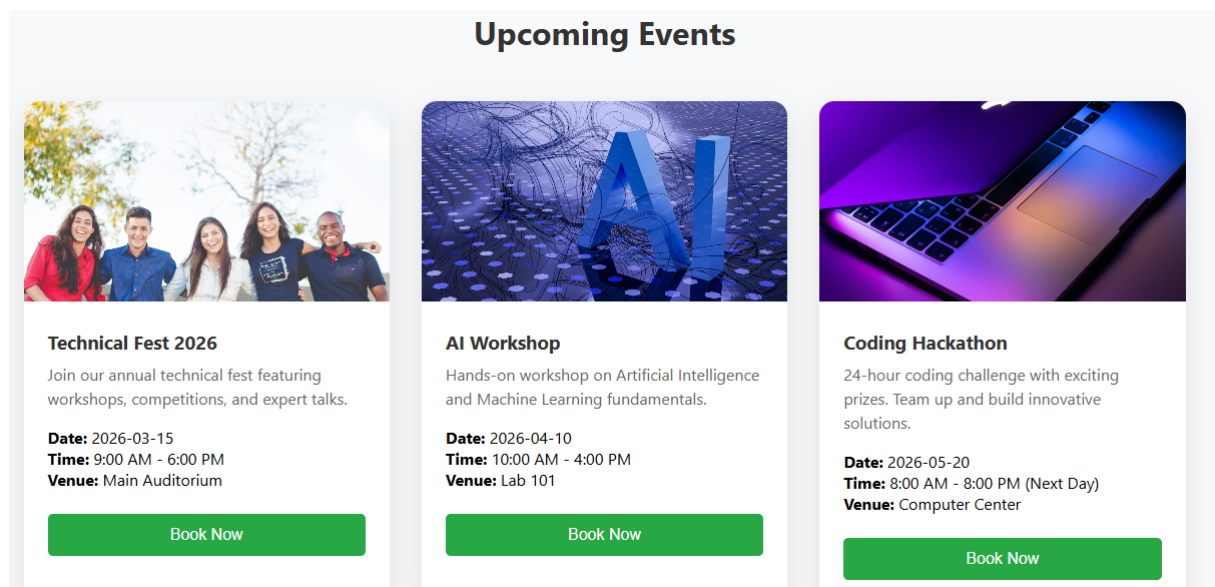


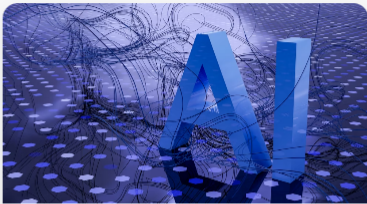
Figure 5.1: User Login Page for Event Booking System



Upcoming Events



Technical Fest 2026
Join our annual technical fest featuring workshops, competitions, and expert talks.
Date: 2026-03-15
Time: 9:00 AM - 6:00 PM
Venue: Main Auditorium
[Book Now](#)



AI Workshop
Hands-on workshop on Artificial Intelligence and Machine Learning fundamentals.
Date: 2026-04-10
Time: 10:00 AM - 4:00 PM
Venue: Lab 101
[Book Now](#)



Coding Hackathon
24-hour coding challenge with exciting prizes. Team up and build innovative solutions.
Date: 2026-05-20
Time: 8:00 AM - 8:00 PM (Next Day)
Venue: Computer Center
[Book Now](#)

Figure 5.2: Display of Upcoming Department Events

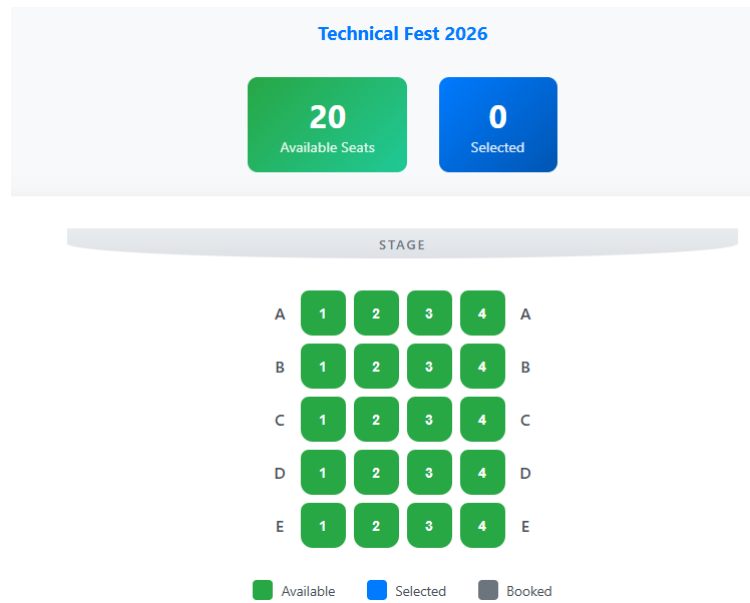


Figure 5.3: Seat Selection Interface for Technical Fest 2026

Customer Details

Full Name
Deepu

Email
deepul@gmail.com

Selected Seats 1 seat(s)

C2

Total Amount ₹500

[Proceed to Payment →](#)

Figure 5.4: Customer Details and Payment Summary Page

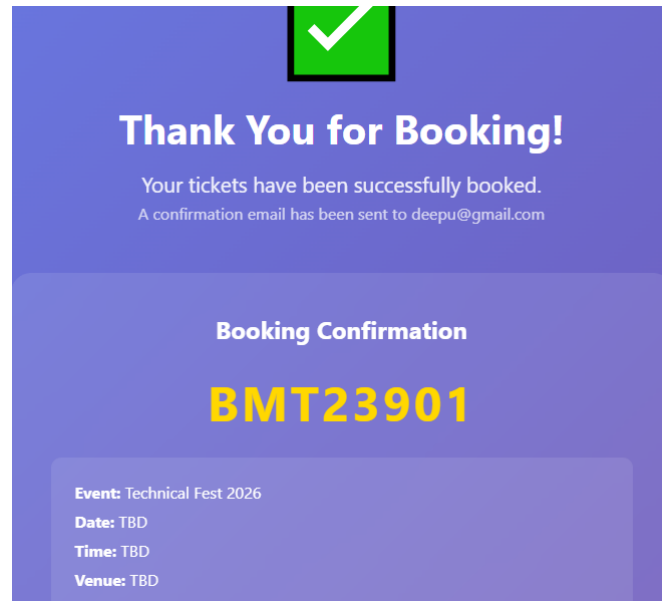


Figure 5.5: Ticket Booking Confirmation Screen

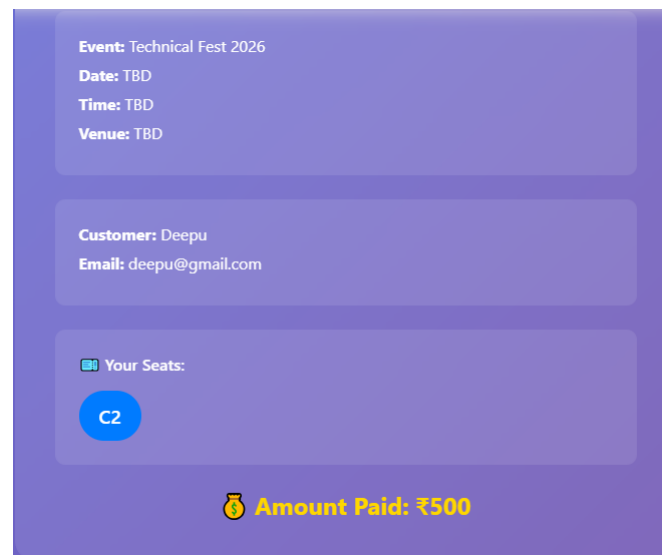


Figure 5.6: Final Ticket Details with Seat and Payment

Chapter 6

CONCLUSION AND FUTURE ENHANCEMENTS

6.1 Conclusion

The Seamless Department Event Booking App successfully provides an efficient and user-friendly solution for managing departmental events and venue bookings. It eliminates the limitations of manual systems by offering real-time availability, conflict-free scheduling, and centralized management.

The application ensures accuracy and transparency through its dynamic booking mechanism and integrated features such as booking confirmation and email notifications. It also demonstrates reliable performance with smooth interaction between system components.

6.2 Future Enhancements

The project can be further enhanced by integrating a backend server and database to store booking information permanently. Additional features such as user authentication, admin dashboard, and multi-event management can be implemented to improve scalability and usability. Online payment gateway integration can be added to support paid event registration. Email or SMS notifications can be included for booking confirmations and reminders. The application can also be extended into a full-stack platform with analytics, attendance tracking, and downloadable digital tickets/QR codes for real-world deployment.

1. BookMyShow

Link: <https://in.bookmyshow.com>

Review Comment: A widely used event and movie ticket booking platform offering real-time seat selection, payment integration, and instant booking confirmation. Suitable for large-scale public events but more complex than required for internal department events.

2. Eventbrite

Link: <https://www.eventbrite.com>

Review Comment: A global event management and ticketing platform used for seminars, conferences, and workshops. Pro-

vides organizer tools, registrations, and ticket sales, but includes advanced features beyond small academic event needs.

3. Ticketmaster

Link: <https://www.ticketmaster.com>

Review Comment: A professional ticket booking system for concerts, sports, and live entertainment. Supports large-scale ticketing and secure booking, but is designed primarily for commercial event management.

Chapter 7

SOURCE CODE

7.1 Source Code

```
1  <!DOCTYPE html>
2  <html>
3  <head>
4  <title>Instant Booking Platform for Department Events</title>
5
6  <style>
7  body {
8      margin: 0;
9      font-family: Arial;
10     background: #eeeeff;
11 }
12 .navbar {
13     background: white;
14     padding: 18px 50px;
15     display: flex;
16     justify-content: space-between;
17     align-items: center;
18 }
19 .logo {
20     color: #5b5cf6;
21     font-size: 24px;
22     font-weight: bold;
23 }
24 .navbar a {
25     margin-left: 20px;
26     text-decoration: none;
27     color: #333;
28     font-weight: bold;
29 }
30 .btn {
31     background: #5b5cf6;
32     color: white;
33     border: none;
34     padding: 10px 22px;
35     border-radius: 6px;
36     cursor: pointer;
37 }
38 .section {
39     display: none;
40     padding: 40px;
41     text-align: center;
```

```

42 }
43 .active {
44     display: block;
45 }
46 .box {
47     background: white;
48     padding: 25px;
49     border-radius: 12px;
50     margin: 20px auto;
51     box-shadow: 0 0 10px #ccc;
52     width: 380px;
53 }
54 input {
55     width: 90%;
56     padding: 12px;
57     margin: 10px;
58     border: 1px solid #ccc;
59     border-radius: 6px;
60 }
61 .hero h1 {
62     color: #5b5cf6;
63     font-size: 40px;
64 }
65 .event-details {
66     text-align: left;
67 }
68 .summary {
69     background: #f2f1ff;
70     padding: 15px;
71     border-radius: 8px;
72     margin-top: 15px;
73 }
74 .error {
75     color: red;
76     font-weight: bold;
77 }
78 .success {
79     color: green;
80     font-weight: bold;
81 }
82 </style>
83 </head>
84
85 <body>
86
87 <div class="navbar">
88     <div class="logo">EventBooking.</div>
89     <div>
90         <a href="#">Home</a>
91         <a href="#">Event Details</a>
92         <a href="#">Book Ticket</a>
93     </div>
94 </div>
95
96 <div class="hero">
97     <h1>Instant Booking Platform for Department Events</h1>
98     <p>Book tickets for seminars, workshops and technical fests easily.</p>
99 </div>
100
101 <div class="box event-details">
102     <h2>Event Details</h2>
103     <p><b>Event Name:</b> Tech Fest 2026</p>
104     <p><b>Department:</b> Computer Science and Engineering</p>
105     <p><b>Date:</b> 25 April 2026</p>
106     <p><b>Time:</b> 10:00 AM</p>
107     <p><b>Venue:</b> Main Auditorium</p>
108     <p><b>Ticket Price:</b> Rs.100</p>

```

```
109     <p><b>Available Tickets:</b> 50</p>
110 </div>
111
112 <div class="box">
113     <h2>Ticket Booking Form</h2>
114     <input type="text" placeholder="Enter Name">
115     <input type="email" placeholder="Enter Email">
116     <input type="text" placeholder="Enter Department">
117     <input type="number" placeholder="Number of Tickets">
118     <button class="btn">Book Ticket</button>
119 </div>
120
121 </body>
122 </html>
```

CHAPTER 8

ADDRESSING COMPLEX ENGINEERING PROBLEM AND SDG

8.1 Mapping of Project to Complex Engineering Problem (CEP)

Level	Attribute	Characteristics	WKs	Justification
WP1	Depth of Knowledge	Requires in-depth engineering knowledge	WK3, WK4, WK5, WK6	Requires Flask, databases, cybersecurity (OTP, SHA256), AI models (LLM, CNN) for secure system.
WP2	Conflicting Requirements	Technical and non-technical conflicts	WK4, WK5, WK6	Balances security and usability with fast transactions and secure authentication.
WP3	Depth of Analysis	Analytical and design thinking	WK3, WK4, WK5	System design and AI integration for pharmacy operations.
WP4	Familiarity	Partially new problems	WK4, WK8	AI chatbot integration introduces modern challenges.
WP5	Applicable Codes	Limited standard solutions	WK6	Custom AI + OTP security design not in standard systems.
WP6	Stakeholder Needs	Multiple stakeholders involved	WK5, WK6, WK7	Supports staff, admin, and customers securely.
WP7	Interdependence	System-level integration	WK3, WK5, WK6	Integrates frontend, backend, AI, and security layers.

Table 8.1: Mapping of Project to Complex Engineering Problem (CEP)

8.2 Mapping of the Project to Complex Engineering Activities(CEA)

EA No.	Attribute	Characteristics	Justification
EA1	Range of Resources for	Use of diverse technologies	Flask, SQLite, AI models, OTP APIs, frontend tech.
EA2	Level of Interactions	Complex module interactions	Billing, AI chatbot, loyalty, security integration.
EA3	Innovation	Modern technologies	AI chatbot + image recognition + OTP security.
EA4	Societal Impact	Impact on users	Improves health-care access and reduces errors.
EA5	Familiarity	Beyond standard practice	AI + pharmacy system integration.

Table 8.2: Mapping of Project to Complex Engineering Activities (CEA)

8.3 SDG Mapping

SDG No.	SDG Title	Relevance to the Project
SDG 3	Good Health and Well-being	AI-based healthcare recommendations and safety improvements.
SDG 9	Industry, Innovation and Infrastructure	Digital transformation using AI and secure systems.
SDG 16	Peace, Justice and Strong Institutions	Secure transactions using OTP and encryption.
SDG 12	Responsible Consumption and Production	Reduces medication errors and improves efficiency.

Table 8.3: SDG Mapping for the Project

References

- [1] Meta Platforms Inc. — <https://react.dev>
- [2] OpenJS Foundation — <https://nodejs.org/en/docs>
- [3] Oracle Corporation — <https://dev.mysql.com/doc>
- [4] BookMyShow Team — <https://bookmyshow.com>
- [5] Eventbrite Inc. — <https://eventbrite.com>
- [6] EventBookings Hub — <https://developers.eventbookings.com>
- [7] Bookwhen — <https://bookwhen.com>
- [8] SuperSaaS — <https://www.supersaas.com>
- [9] Momentus — <https://gomomentus.com>
- [10] Stripe — <https://stripe.com/docs>